MOHAMED HAJBI

Software Engineer

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Software Engineer with 2.5+ years of experience building Python backends and REST APIs, specialized in FastAPI and PostgreSQL/PostGIS for large-scale geospatial data, with API response times cut by ~80%. Full-stack background across Django, Spring Boot, NestJS, Vue.js, React and Angular, with JWT security, Docker and automated CI/CD deployments on AWS. Engineering degree in Computer Science.

PROFESSIONAL EXPERIENCE

Software Engineer | Climate Innov Digital *September 2025 – Present*

- Developed and optimized FastAPI REST endpoints for large geospatial datasets, reducing API response times by approximately 80%.
- Designed and implemented the PostgreSQL/PostGIS database architecture for large-scale geospatial data storage and querying.
- Optimized database performance through index design and query optimization.
- Built end-to-end pipelines integrating multiple models into the application, from input data to results served by the API.
- Containerized the models with Docker and orchestrated their asynchronous execution with Celery task queues and RabbitMQ.
- Assessed the compute needs of each workload to choose between local execution and cloud instances, and sized resources accordingly.
- Implemented secure authentication and authorization using JWT.
- Automated deployment pipelines using GitHub Actions and AWS Systems Manager (SSM).
- Implemented AWS deployment infrastructure using S3, EC2, ECR and Elastic IP.

Software Engineer Intern | Climate Innov Digital *February 2024 – August 2025*

- Developed RESTful APIs using FastAPI for geospatial data processing and retrieval.
- Contributed to PostgreSQL/PostGIS database design and large-scale geospatial data management.
- Processed, cleaned and transformed geospatial datasets for analysis.
- Built Docker-based environments for isolating backend and geospatial processing services.

Backend Developer Intern | ASM *July 2024 – August 2024*

- Developed backend functionality for a messaging application using Spring Boot, Firebase and MongoDB.
- Integrated Firebase Cloud Messaging (FCM) and EMQX MQTT.

Full-Stack Developer Intern | TEKABDEV *July 2023 – August 2023*

- Built an admin dashboard and a 404 page using Vue.js.
- Developed an order-tracking API using NestJS, with PostgreSQL and Docker.

Full-Stack Developer Intern | Ahwa Solution *February 2022 – May 2022*

- Designed and developed a gym management web platform using Django.
- Integrated the Stripe payment API and implemented real-time notifications.

PROJECTS

- **Sales Management System**: Microservices architecture with Spring Boot, .NET and Angular, REST APIs for sales operations and analytics, JWT security, API Gateway and Eureka service discovery.
- **Voice-Controlled News App**: React.js news platform with hands-free navigation using Alan AI speech recognition.

EDUCATION

Engineering Degree in Computer Science
Polytechnique Sfax (IPSAS) | 2022 – 2025

Bachelor's Degree in Information Technology
ISET Sidi Bouzid | 2018 – 2022

SKILLS

- **Languages:** Python, JavaScript, Java, HTML, CSS
- **Backend:** FastAPI, Django, Flask, Spring Boot, NestJS, RESTful API design, JWT, Celery, RabbitMQ, microservices
- **Frontend:** ReactJS, Vue.js, Angular
- **Databases:** PostgreSQL, PostGIS, MongoDB, Firebase, indexing and query optimization
- **Cloud & DevOps:** AWS (S3, EC2, ECR, Elastic IP, SSM), Docker, GitHub Actions, CI/CD, Git

LANGUAGES

English (B2) · French (intermediate, working level)